Scalability & Future Plan

Date	29 September 2026
Team ID	SWTID-2026-3976
Project Name	AI FAQ Assistant API (AnswerBot)
Maximum Marks	1 Mark

Scalability & Future Plan:

This document explains how the solution can grow to serve more users and data, and the roadmap for future improvement.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	No front-end user interface	Non-technical users cannot use the API directly	High
2	Keyword search only (no semantic search)	Related but differently worded questions may be missed	High
3	Admin and editor roles, likes and feedback not implemented	Limited moderation and quality signals	Medium
4	No rate limiting or load testing yet	AI endpoints could be overused; capacity unknown	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single Node.js instance, stateless JWT	Run multiple instances behind a load balancer (Nginx / cloud) or containers
2	Data Storage	MongoDB Atlas single database	Add indexes on question/answer/category; use replica sets and sharding when data grows
3	Performance	Direct database and Gemini calls	Cache popular searches (Redis) and queue heavy AI jobs
4	Security	JWT, bcrypt, validation	Add rate limiting, Helmet, refresh tokens and role-based access control

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Web front end, admin and editor roles, category endpoints	Next 1 month	Usable product for support teams
Phase 3	Semantic search with embeddings, likes and answer feedback, tags	Next 2-3 months	Better answers and quality signals
Phase 4	Analytics dashboard, Docker/cloud deployment, CI/CD, multilingual FAQs	Next 4-6 months	Production-ready and scalable